

OTHER GRADE PAPER

Evaluation of grading of recommendations assessment, development, and evaluation approach adherence in clinical practice guidelines: a systematic survey of criteria and their applications

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Abstract

Objectives: Many clinical practice guidelines (CPGs) claim to follow the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach, yet their actual adherence remains uncertain. We aimed to identify and evaluate existing criteria for evaluating adherence to GRADE in CPGs and describe the extent of adherence revealed by their application.

Study Design and Setting: We systematically searched PubMed, Embase, Web of Science, and the Cochrane Library from January 2004 to November 2024, supplemented by manual reviews of the GRADE handbook and related websites. Eligible documents were those that either proposed or applied criteria for evaluating GRADE adherence. We summarized the criteria offered, evaluated the extent to which they meet standards for a structured instrument, and analyzed adherence findings.

Results: We identified 11 eligible documents: eight proposed evaluation criteria (four applying their own to evaluate CPGs) and three applied existing criteria. From these criteria, we derived 10 clearly discrete items that can be used to define good adherence to GRADE. The most common items addressed the criteria for assessing certainty of evidence (CoE) and criteria for making recommendations (seven of eight documents), while question framing and outcome selection and importance rating were less frequent (3/8). Among the eight documents proposing criteria, four failed to report their development methods and response options; furthermore, none conducted user testing or provided guidance manuals. The number of CPGs assessed in the included studies ranged from 4 to 86. Reported adherence to GRADE varied considerably across these studies; nevertheless, the median adherence rates were suboptimal for five of the 10 items.

Conclusion: Existing criteria for assessing GRADE adherence are varied and suboptimal in usability. Evaluations based on these criteria reveal significant application gaps in key areas among CPGs that claim to use GRADE. These findings highlight the need to develop a structured, user-tested evaluation instrument and improve the implementation of GRADE in CPGs. © 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

Keywords: Guidelines; GRADE; Systematic review; Guideline adherence; Criteria; Instrument

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Plain Language Summary

Many medical guidelines say that they use a trusted scientific method called GRADE to make their advice reliable. But we neither know if they actually follow it correctly nor were there good tools to check. We wanted to find out: 1) what checklists exist to audit GRADE use? and 2) what do those checklists tell us about how well guidelines really apply GRADE?

We searched scientific databases and official GRADE sources for studies that either created or used checklists to evaluate GRADE in guidelines. We reviewed these checklists, checked their quality, and summarized what they found when applied.

We found 11 relevant studies. The checklists they proposed were inconsistent and poorly designed; none were tested with real users or came with instructions. When used, they showed that many guidelines fail to properly apply key steps of the GRADE method, even though they claim to use it. For half of the important steps, less than 50% of guidelines followed them correctly.

This means that current tools for checking GRADE use are unreliable, and many guidelines may not be as trustworthy as they claim. To protect patients and improve medical advice, we urgently need a better, user-friendly audit tool and more support to help guideline developers apply the GRADE method correctly.

1. Introduction

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach provides a widely recognized and accepted guidance for helping develop high-quality clinical practice guidelines (CPGs) [1–3]. The GRADE approach provides a systematic and transparent framework for assessing the certainty of evidence (CoE) and moving from evidence to recommendations. More than 120 organizations from 19 countries have endorsed or are using GRADE [4].

Despite its rigorous methodology, the consistent and appropriate application of the GRADE approach remains a challenge. In 2020, reviews of CPGs in the United States and Australia found that, although these CPGs claimed to use the GRADE approach, serious limitations regarding CoE assessment, evidence summarization, and the application of evidence to decision (EtD) criteria were common [5,6]. These limitations can potentially undermine guideline trustworthiness, thus compromising their usefulness in clinical decision-making.

When assessing the quality of CPGs, standardized quality assessment tools, such as the Appraisal of Guidelines for Research and Evaluation (AGREE) II and the National Guideline Clearinghouse Extent of Adherence to Trustworthy Standards (NEATS) instrument (and its modified version), have proved useful [7–9]. However, these existing instruments fail to capture key elements of the GRADE approach [10]. Therefore, a dedicated instrument for reliably evaluating GRADE adherence in CPGs is needed.

In light of these considerations, this systematic survey aims to 1) identify existing criteria for addressing adherence to the GRADE approach in CPGs; 2) assess whether the included documents that suggested these criteria were ideal instruments based on our customized standards; and 3) summarize findings from applications of these criteria

within a critical context. Through this work, we hope to contribute to ongoing efforts to enhance the quality of CPGs.

2. Methods

2.1. Data sources and searches

Our reporting followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (Appendix 1) but was not registered. We searched PubMed, Embase, Web of Science, and the Cochrane Library (Cochrane Database of Systematic Reviews and Cochrane Central Register of Controlled Trials) from January 1, 2004, to November 26, 2024. We developed the search strategy in collaboration with an experienced research librarian. To identify other potentially eligible documents, we manually reviewed the GRADE handbook, the websites of GRADE working group (GWG) and GRADE centers/networks, and the reference lists of included documents (see Appendix 2 for the full search strategy).

2.2. Document selection

Pairs of reviewers independently screened titles and abstracts and then full texts. They resolved discrepancies by consensus or involving a third reviewer if necessary.

Studies were eligible if they 1) suggested criteria (or synonyms such as requirements, standards, instruments, tools, or checklists) for evaluating adherence of CPGs to the GRADE approach or 2) applied such criteria to evaluate CPGs' adherence to the GRADE approach. We did not apply any language restrictions. However, we excluded publications that were unavailable in full text or evaluated GRADE application but did not specify the assessment items.

What is new?**Key findings**

- We identified eight documents proposing criteria to evaluate adherence to the GRADE approach in CPGs and identified 10 clearly discrete items.
- Half of the criteria failed to provide response options, and none of them undertook user testing or provided manuals. Therefore, a satisfactory structured instrument for evaluating GRADE adherence in CPGs is currently lacking.
- The actual implementation of the GRADE approach in CPGs remains suboptimal overall, as evaluations have shown major nonadherence in key areas such as question formulation, outcome selection and prioritization, criteria for CoE assessment, evidence summary, and recommendation making.

What this adds to what is known?

- This is the first systematic survey to report on existing criteria for evaluating adherence to GRADE approach in CPGs and their applications.
- While the GRADE approach has been widely adopted, the lack of a standardized instrument to evaluate its adherence has remained an understudied critical gap.

What is the implication and what should change now?

- Our findings highlight the critical need to prioritize both the implementation of GRADE and its practical evaluation.
- Our empirical evidence, including the summary of evaluation items and item-level adherence results, can inform the development of an idea instrument for evaluating GRADE adherence in CPGs. This could enable the field to monitor implementation, identify recurring challenges, and report GRADE adherence transparently.

2.3. Data extraction

We used separate, standardized data extraction forms: one for documents suggesting criteria and another for those applying criteria. One reviewer extracted data, and a second reviewer checked.

For documents suggesting criteria, we extracted basic information (title, journal/source, authors, year, and publication type); development process; items; response options; and availability of guidance manuals.

For documents applying criteria, we extracted basic information (title, journal/source, authors, and year); criteria applied and their source; characteristics of evaluated CPGs (source, region, publication year, and number); and adherence results (frequency and percentage of CPGs adhering to each item).

2.4. Development and application of standards for an ideal structured instrument

To inform standards for appraising documents that proposed GRADE criteria, we examined formal instruments for addressing risk of bias [11–14], systematic review quality [15,16], guideline trustworthiness [8,17,18], and subgroup analysis credibility [19] and consulted the framework for developing quality assessment tools [20]. Based on this review and through iterative team discussions and consensus, we developed the following standards for an ideal structured instrument to evaluate GRADE adherence: 1) reports its development methods, particularly the criteria for item selection; 2) provides response options for each item; 3) provides a user guidance manual; and 4) has undergone user testing.

2.5. Data synthesis and analysis

We extracted each suggested criterion verbatim from the source documents and then judged whether different criteria expressed the same essential concept. Some judgments regarding underlying meaning proved clear. For example, “CoE,” “quality of evidence,” and “confidence in the estimates” are conceptually equivalent. In such cases, we chose a single presentation format, here, labeled as “CoE”. The items varied considerably in their granularity and presentation, making judgments about conceptual equivalence challenging. Therefore, we prepared both a restricted number of items that we judged discrete and a larger number with some overlap. We generated descriptive summaries of extracted data and presented results in a combination of summary tables and narrative descriptions.

For studies assessing guideline adherence, we performed descriptive statistics using the median and range of guideline adherence rate (defined as the percentage of CPGs that adhered to each item among the evaluated CPGs; in instances where adherence was not reported as a binary outcome, relevant categories were merged to derive a comparable measure, with details provided in the results section), whenever this rate was reported by three or more studies.

3. Results**3.1. Search results**

Database searches identified 10,581 records (Fig. 1). After removing duplicates, of 7568 remaining, we screened

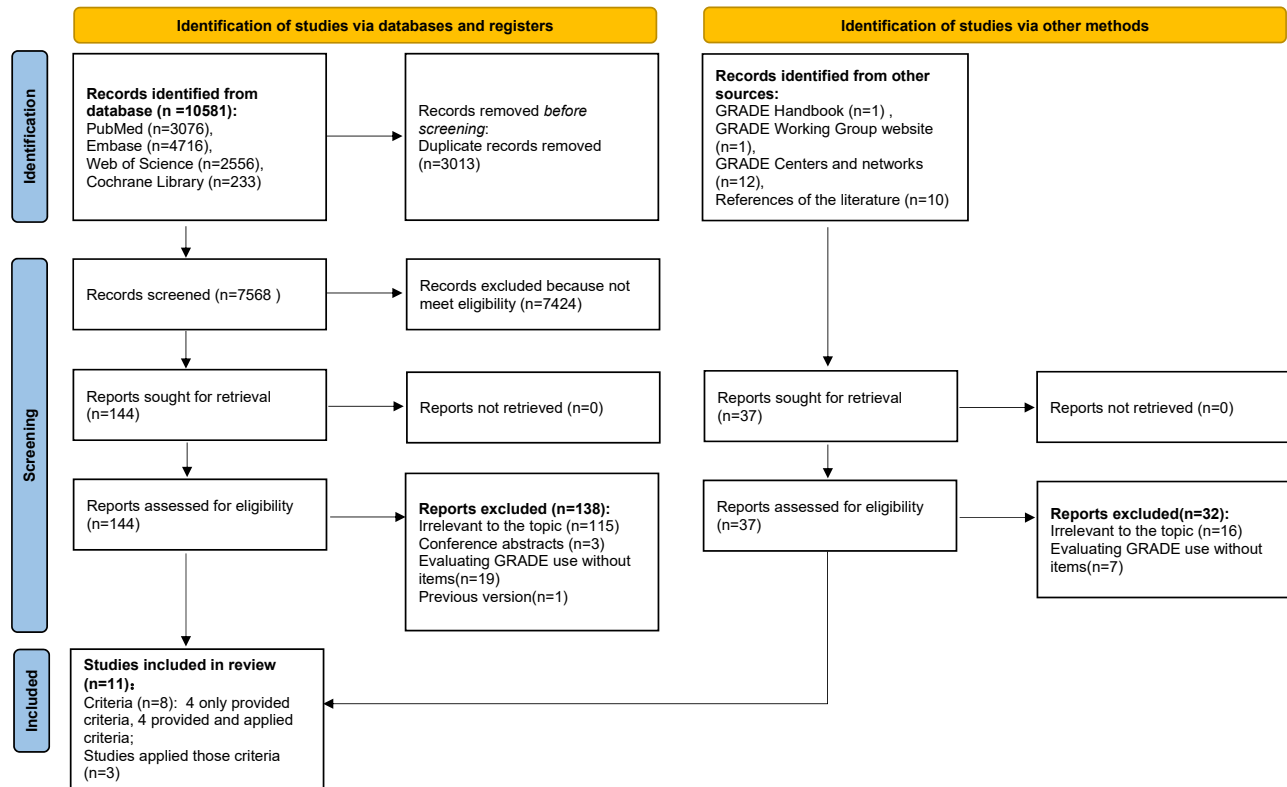


Figure. PRISMA flowchart of study selection and inclusion process. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

144 full texts, of which six proved eligible. After adding five records from other sources, 11 were included: eight proposed adherence criteria (four also applied them) and three applied existing criteria to evaluate GRADE adherence in CPGs.

3.2. Criteria for assessing GRADE adherence

Appendix 3 presents characteristics of the eight included documents [6,10,21–26]. Appendix 4 provides the verbatim criteria from the original documents.

The GRADE handbook was the first to provide criteria for assessing adherence to the GRADE approach [21]. These criteria were updated in 2016 on the GWG website [22]. The GRADE guidance group (GGG) published a set of seven minimal requirements for stating GRADE use: four for CoE assessment and three for recommendation making [23].

The Australian GRADE center developed an 18-question checklist, grouped in seven categories, for assessing GRADE adherence in CPGs [6]. Colunga-Lozano et al. proposed three criteria: reporting patient, intervention, comparator, and outcome (PICO) questions, evidence summary, and CoE rating factors [24]. Norling B et al. and Murray R et al. designed data abstraction forms to capture the application of GRADE and EtD framework in CPGs [25,26]. Hazlewood GS et al. determined the alignment

between GRADE and AGREE II, specifying which GRADE steps are covered and which are not [10].

Table 1 presents a succinct fully aggregated categorization of 10 items derived from the criteria-suggesting documents, showing their sources. Appendix 5 presents a less aggregated categorization listing 22 items for reference.

Criteria for assessing CoE and making recommendations were the most common items, each appearing in seven of the eight documents. Six of eight documents mentioned CoE categories and evidence summary; five mentioned recommendation terminology and judgment reporting; and four mentioned definitions. Framing PICO questions, selecting and rating outcomes, and reporting GRADE in abstract were less common.

Regarding criteria for moving from evidence to recommendations, the GGG's recent update removed "resource use" from the basic EtD criteria, grouping it with equity, acceptability, and feasibility as context-dependent criteria to be applied according to decision type or group mandate [23]. In addition, two documents [25,26] mentioned equity, acceptability, and feasibility, while only one mentioned problem priority [17].

Table 2 presents our four standards for an ideal structured evaluation instrument and the extent to which the eight criteria-suggesting documents met these standards. Of the eight documents, four failed to report their

Table 1. A succinct fully aggregated categorization of 10 items included in the criteria-suggesting documents

Items	Documents							
	GRADE handbook (2013) [21]	GRADE working group (2016) [22]	Barker TH et al. (2021) [6]	Schünemann HJ et al. (2023) [23]	Colunga-Lozano LE et al. (2021) [24]	Norling B et al. (2024) [25]	Murray R et al. (2023) [26] ^a	Hazlewood GS et al. (2014) [10]
Reporting GRADE in abstract						✓		
Definitions of certainty of evidence, the ranking of certainty, or each category of the strength of recommendations	✓	✓	✓	✓				
Framing PICO questions			✓		✓			✓ ^b
Selecting and rating the importance of outcomes			✓ ^c			✓ ^d		✓ ^c
Criteria for assessing the certainty of evidence	✓	✓	✓	✓	✓	✓		✓
Certainty of evidence categories	✓	✓	✓	✓		✓		✓
Summarizing the evidence	✓	✓	✓	✓	✓	✓		
Criteria for making recommendations	✓ ^e	✓	✓ ^e	✓ ^f		✓ ^g	✓ ^h	✓ ⁱ
Strength/direction of recommendation terminology	✓	✓	✓	✓				✓
Reporting of judgments	✓	✓	✓	✓				✓

GRADE, Grading of Recommendations Assessment, Development, and Evaluation; PICO, patient, intervention, comparator, and outcome.

^a This document only focused on GRADE evidence to decision.

^b The item stated “the health question(s) covered by the guideline is (are) specifically described,” not specifying PICO.

^c Rating the importance of outcomes.

^d Description of how outcomes selected.

^e Requiring four criteria: balance of desirable and undesirable consequences, certainty of evidence, values and preferences of those affected, and resource use.

^f Distinguishing EtD criteria: Guideline developers should describe the magnitude and balance of desirable and undesirable health effects of the options that are considered, the value of each outcome, and the certainty of the evidence for the aforementioned EtD criteria. Other EtD criteria, such as resource use, equity, acceptability, and feasibility that are used, may vary, depending on the type of recommendation or decision, and the mandate of the group that is making the recommendation or decision.

^g Mentioning seven criteria: overall CoE; benefits and harms; values and preferences; resource use; acceptability; equity; and feasibility.

^h Mentioning nine criteria: problem priority, benefits and harms, certainty of the evidence, outcome importance, balance, resource use, equity, acceptability, and feasibility.

ⁱ Mentioning four criteria: health benefits, side effects, and risks; patients' values/preferences; overall strength/quality of evidence; and resource implications.

Table 2. Judgments regarding whether the eight criteria-suggesting documents met our standards for an ideal structured evaluation instrument

Standards	Documents							
	GRADE handbook (2013) [21]	GRADE working group (2016) [22]	Barker TH et al. (2021) [6]	Schünemann HJ et al. (2023) [23]	Colunga-Lozano LE et al. (2021) [24]	Norling B et al. (2024) [25]	Murray R et al. (2023) [26] ^a	Hazlewood GS et al. (2014) [10]
Reporting development methods, particularly criteria for item selection	×	×	✓	×	×	✓	✓	✓
Providing response options for each item	×	×	✓	×	×	✓	✓	✓
Providing guidance manuals	×	×	×	×	×	×	×	×
Conducting user testing	×	×	×	×	×	×	×	×
Final judgments	×	×	×	×	×	×	×	×

GRADE, Grading of Recommendations Assessment, Development, and Evaluation.

^a Piloting among the author team but not the target users.

development methods, four lacked response options, and none offered a guidance manual or carried out user testing.

3.3. Results of GRADE adherence assessment in CPGs

Table 3 presents characteristics and results from the seven studies that evaluated GRADE adherence in CPGs. Four of these studies were also involved in proposing the criteria summarized in Table 1 [6,10,24,25]. The other three [5,27,28] applied the existing criteria: two [5,28] used the GGG's 2016 update [22] and one [27] used criteria from the Australian GRADE center [6]. Most studies were published in 2020 or later. These studies included between four and 86 CPGs each, and the evaluated CPGs spanned from 2000 to 2023 and covered diagnostic, therapeutic, and prognostic topics.

One study evaluated GRADE reporting in abstract, showing high adherence (89.1%). Overall, most CPGs used definitions of key terms consistent with GRADE (median adherence rate: 74%, range: 5% to 89.6%), used GRADE CoE categories (median adherence rate: 55.1%, range: 5% to 98.5%), provided the direction and strength of recommendations (median adherence rate: 53.7%, range: 40.9% to 100%), and reported the related judgments (median adherence rate: 53%, range: 5% to 100%). Fewer than half of the CPGs framed PICO questions (median adherence rate: 40%, range: 0% to 100%), reported outcome selection and importance rating (median adherence rate: 41.5%, range: 2% to 82.6%), explicitly considered criteria for CoE assessment (median adherence rate: 42.6%, range: 0% to 75%), provided evidence summaries (median adherence rate: 39.3%, range: 4% to 53.7%), or considered GRADE criteria for making recommendations (median adherence rate: 36.4%, range: 2% to 100%).

4. Discussion

We have systematically reviewed existing criteria for evaluating adherence to GRADE approach in CPGs and their application. We identified eight criteria-suggesting documents and derived 10 clearly discrete items (Table 1). None of the criteria-suggesting documents met our customized evaluation standards for a satisfactory structured evaluation instrument (Table 2). Preliminary evaluations of adherence to existing criteria reported major limitations of GRADE application in the existing CPGs (Table 3).

4.1. Evolution and comparison of the current criteria

The criteria for assessing GRADE adherence in CPGs vary across documents. Both criteria for assessing CoE and making recommendations proved the most common items, likely representing GRADE's core functionality [29–31]. Fewer documents included PICO framing, outcome selection, and ranking, possibly because some

Table 3. Characteristics and GRADE adherence of the evaluated CPGs in the seven studies

Characteristics and results	Documents						
	Barker TH et al. (2021) [6]	Colunga-Lozano LE et al. (2021) [24]	Norling B et al. (2024) [25]	Hazlewood GS et al. (2014) [10] ^j	Dixon C et al. (2020) [5]	Chen Z et al. (2022) [27]	Dahm P et al. (2024) [28]
Characteristics of the evaluated CPGs							
Source	Australian National Health and Medical Research Council	National Healthcare Technology Excellence Center	Eastern Association for the Surgery of Trauma	Rheumatoid arthritis guidelines	National Guidelines Clearing House	Pressure injury prevention guidelines	European Association of Urology
Region	Australia	Mexico	United States	Global	United States	Global	Europe
Publication year	2011 to 2018	2015 to 2017	2014 to 2022	2000 to 2012	2011 to 2018	2015 to 2020	2023
Number of the evaluated CPGs	15	86	46	44	67	4	20
GRADE adherence for each item, n (%)							
Reporting GRADE in abstract	NR	NR	41 (89.1%)	NR	NR	NR	NR
Definitions of certainty of evidence, the ranking of certainty, or each category of the strength of recommendations	11 (73%)	NR	NR	NR	60 (89.6%) ^a	3 (75%)	1 (5%)
Framing PICO questions	6 (40%)	0 (0%)	NR	NR ^h	NR	4 (100%)	NR
Selecting and rating the importance of outcomes	5 (33%) ^b	NR	38 (82.6%) ^c	1 (2%) ^b	NR	2 (50%) ^b	NR
Criteria for assessing the certainty of evidence	9 (60%)	0 (0%)	35 (76.1%), 4 (9.1%) ^d	22 (50%) ^e	9 (13.4%)	3 (75%)	1 (5%)
Certainty of evidence categories	10 (67%)	NR	13 (28%)	19 (43.2%)	66 (98.5%)	3 (75%)	1 (5%)
Summarizing the evidence	8 (53%)	1 (4%)	77 (55.8%), 2 (1.4%) ^f	NR	36 (53.7%)	2 (50%)	1 (5%)
Criteria for making recommendations	4 (27%)	NR	21 (45.7%)	1 (2%) ^g	36 (53.7%)	4 (100%)	1 (5%)
Strength/direction of recommendation terminology	8 (53%)	NR	NR	18 (40.9%)	36 (53.7%)	3 (75%)	20 (100%)
Reporting of judgments	8 (53%)	NR	NR	NR ⁱ	NR	4 (100%)	1 (5%)

GRADE, Grading of Recommendations Assessment, Development, and Evaluation; CPGs, clinical practice guidelines; NR, not reported; PICO, patient, intervention, comparator, and outcome. Footnotes clarify variations in how items were defined or reported across the included studies.

^a Definitions of certainty of evidence.

^b Rating the importance of outcomes.

^c Description of how outcomes selected.

^d 35 (76.1%) for rating down certainty of evidence, four (9.1%) for rating up certainty of evidence.

^e At least one criterion for assessing the certainty of evidence.

^f 77 (55.8%) for evidence profiles, two (1.4%) for summary of findings tables (note: questions as the unit).

^g The criterion of “values or preferences of patients.”

^h This study did not provide evaluation data for this item while proposed it.

ⁱ This study did not provide quantifiable evaluation data for this item, describing it only as “rarely explicit.”

^j For this study, we used the addressed (including “inconsistently addressed” and “consistently addressed”) frequency and percentages as the results.

researchers viewed these as general requirements for CPG development, independent of GRADE use [32].

The variation in requirements for EtD criteria arises from the context-specific nature of “resources and costs.” For public policy decision-making, considerations such as resource use, equity, acceptability, and feasibility are particularly critical [31,33].

Inconsistency in evaluation criteria for claiming GRADE use risks exacerbating variability in GRADE implementation in CPGs, ultimately undermining its intended role in enhancing guideline quality.

4.2. Shortcomings of existing criteria-suggesting documents as operational instruments

Existing criteria-suggesting documents fall short of being satisfactory operational instruments for evaluating CPGs’ adherence to the GRADE approach. All these documents were developed without user testing and lacked guidance manuals; some documents even failed to specify response options or item selection criteria, compromising their content quality, presentation, and usability. Notably, Colunga-Lozano LE et al. likewise noted the absence of a standardized, validated tool, acknowledging that their own ad hoc instrument remained untested [24]. Furthermore, only a limited number of studies have applied these proposed criteria for evaluation since their publication.

To ensure that “using GRADE” is more than a label, its implementation must be rigorously evaluated, just as quality assessment tools are used to evaluate and enhance the quality of research [8,15–18].

4.3. Current status of adherence to the GRADE approach in CPGs

Despite the instrument limitations, our findings reveal consistent and clear patterns of nonadherence across multiple CPGs. While CPGs aligned well with GRADE’s definitions, CoE categories, and terminology for the strength/direction of recommendations, performance was poorer in question formulation, outcome selection and prioritization, evidence summary, criteria for CoE assessment, and recommendation making. A key and disconcerting finding is that less than half of the CPGs explicitly reported the PICO questions. It suggests that even basic, critical steps in the guideline development process are being overlooked or under-reported, which has significant implications for the validity of the guidelines produced. Areas where we found high adherence are likely to be well-specified in the GRADE approach itself, whereas areas of low adherence may point to both genuine challenges for guideline developers and ambiguities in the assessment instruments. This may be because these areas require greater methodological sophistication and more resource investment.

The studies that reported these adherence findings did not systematically investigate the underlying causes.

However, they did offer several plausible inferences, including users’ significant misunderstandings of GRADE (eg, viewing it as only a minor component of guideline development), perceived burden of full implementation, methodological complexity requiring specific training and expertise, a lag in adopting newer methodology, inadequate reporting or lack of transparency, subjective influences (eg, panel’ skepticism toward weak recommendations), perceived irrelevance of certain GRADE steps in specific contexts, and the use of alternative frameworks [6,10,25,27,28].

A recent qualitative study exploring the challenges faced by systematic review authors in applying the GRADE approach identified similar barriers, including insufficient knowledge and training, time constraints, complexity of the system, motivational and attitudinal challenges, and financial limitations [34]. Guideline developers likely face identical obstacles. An alternative explanation for our findings is reporting bias; some GRADE steps may have been undertaken but not fully reported in the final guidelines, leading to an underestimation of actual adherence. Addressing them therefore requires a multifaceted strategy, such as enhanced training and organizational support, transparent reporting, and interorganizational collaboration to share the burden [28,34].

Although guidelines using GRADE tend to score higher on quality appraisals (eg, AGREE II) [2,6,35], this benefit hinges on its faithful application. Nonadherence to GRADE can reintroduce the biases it aims to eliminate, thereby undermining the foundational purpose of evidence-based guideline development. It is also crucial to recognize that the quality of the underlying evidence synthesis (eg, meta-analysis) is a more foundational determinant of the ultimate recommendation’s validity. Some studies have also reported that systematic reviews frequently fail to adhere to established criteria for applying GRADE [36–38].

When guideline development—a resource-intensive process—fails in these aspects, end users may either lose confidence in the recommendations or inappropriately adopt unreliable recommendations, resulting in waste of resources or diminished quality of clinical decision-making [6,39].

4.4. Strengths and limitations

This is the first systematic survey to report on existing criteria for evaluating adherence to GRADE approach in CPGs and their applications. Summarizing the items of current criteria provides candidate items and references for future instrument development. Furthermore, the identified gaps in both adherence evaluation and GRADE implementation itself provide critical insights for future improvement.

Our study has some limitations. First, synthesizing criteria with varied information density into the final list of items involved subjective judgments, including decisions

regarding lumping or splitting items. For example, others may split outcome selection and outcome importance rating as separate items. To address this, we have provided both aggregated and disaggregated versions of the item categorization in [Appendix 5](#). Second, due to the limitations of current measurement tools, there is likely significant heterogeneity and potential misclassification in how different assessors apply them. Therefore, our findings represent adherence as measured by these imperfect instruments, a snapshot, not a definitive measure of “true” adherence. Once a robust assessment tool is available, large-scale surveys of CPGs will be required to more accurately understand their adherence to GRADE. Third, our study focuses exclusively on CPGs’ adherence to GRADE. While GRADE is widely endorsed, it is neither the sole mechanism for ensuring CPG quality nor is it a comprehensive guarantee of trustworthiness. A comprehensive evaluation of guideline quality requires these broader considerations.

4.5. Reflections and future research

A structured and user-tested instrument with manuals should be developed to evaluate GRADE adherence, serving as a supplement to existing guideline appraisal instruments like AGREE II. These tools can be used independently or in combination, depending on the specific evaluation objective. Although the GRADE approach is still being updated and refined, the definition of core GRADE elements [40] may facilitate the development of a new instrument that addresses GRADE essentials. The core GRADE series, which presented an overview of the essential elements of the GRADE approach, can thus serve as an updated reference for identifying essential items in evaluation criteria development [29,41].

5. Conclusion

This systematic survey reviewed current criteria for evaluating GRADE adherence in CPGs. We found that these criteria are not fully consistent, and that the documents proposing them suffer from operational limitations, such as a lack of user testing and guidance manuals. Despite widespread claims of GRADE use in CPGs, assessments reveal significant shortcomings in its application, including in PICO question formulation, selection and importance rating of outcomes, evidence summaries, criteria for accessing CoE, and criteria for making recommendations. These findings provide an empirical foundation for standardizing GRADE evaluation practices. Future efforts should focus on improving the evaluation of GRADE application, such as developing a structured and user-friendly standardized instrument and implementing supportive strategies to bridge the gap between GRADE principles and their practical application.

CRedit authorship contribution statement

Siyu Yan: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Gordon H. Guyatt:** Writing – review & editing, Methodology, Conceptualization. **Xiangying Ren:** Writing – review & editing, Investigation, Data curation. **Pablo Alonso-Coello:** Writing – review & editing, Methodology, Conceptualization. **Philipp Dahm:** Writing – review & editing, Methodology, Conceptualization. **Qiao Huang:** Writing – review & editing, Conceptualization. **Yongbo Wang:** Writing – review & editing, Conceptualization. **Xiaomei Yao:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Yinghui Jin:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

SY, GHG, PA-C, PD, XY, and YJ are members of the GRADE working group. There are no competing interests for any other author.

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Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112186>.

Data availability

Datasets supporting the conclusions of this article are available upon request from the first author and corresponding author.

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